B.O.T.A.K. // STAGE 2 // SYSTEM ONLINE

TEST BANK

STAGE 2 · V1.0 · NOV 2026

Algorithms, Data Structures & Connectionist AI



40 TRUE OR FALSE PART A ›	**42** MULTIPLE CHOICE PART B ›	**13** OPEN-ENDED PART C ›

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MODULE	Algorithms, Data Structures & Connectionist AI
VERSION	Test Bank V1.0 · Stage 2 · November 2026 · companion to Study Guide V1.0
CONTENTS	Part A: 40 true/false · Part B: 42 multiple choice · Part C: 13 open-ended · Answer key (after Part C)

PILOT ID · type your details

PILOT NAME

SQUAD / CLASS

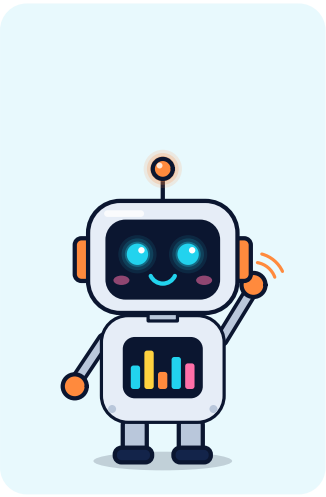
LAUNCH DATE

Best in Adobe Acrobat Reader: fill-in boxes, auto-scoring and word counters.

START MISSION ▶

Mission briefing

How to use this Test Bank, and how this interactive PDF works.



HOW TO USE THIS TEST BANK

Topic coverage follows the Study Guide: algorithms (M1–M4), data structures (M5–M7) and connectionist AI (M8–M12). Each question is labelled with the Study Guide mission it tests (M1 = Mission 1, and so on). Answer all questions first, then check the answer key. Every answer is explained, so read the explanation even when you were right.

Several questions reuse examples from the guide's labs (the six score cards, the maze, the A–F tree, the weighted S–G graph, the coin-change problem, the book stack, the binary search tree, the friendship network and the corridor robot). Everything you need is in the question; no internet access is required.

YOUR CONTROLS

Tap to answer

Each true/false card has TRUE and FALSE buttons. Multiple-choice cards have a button for every option (square boxes when you may pick several).

Type your answers

Use the "fix it" line on false statements and the big answer boxes in Part C. A live counter tracks your words towards the 100-word target.

Run a system check

At the end of Parts A and B, press RUN CHECK: every card gets a green ✓ or red ✗ and your score appears. Mission Control then ranks you.

Jump around

Tap a mission on the map, MISSION MAP in the header, or KEY › on any card to see its explanation. ‹ BACK in the key returns you to the question.

PILOT TIP

Answer boxes work in any PDF viewer. Auto-scoring, ✓/✗ marks and word counters need **Adobe Acrobat Reader** (free). In other viewers, mark yourself with the answer key.

NEXT: OPEN THE MISSION MAP

MISSION MAP ›

Mission map

Twelve Study Guide missions. Tap a T/F, MC or OPEN chip to jump straight to those questions.

ALGORITHMS M1–M4 DATA STRUCTURES M5–M7 CONNECTIONIST AI M8–M12

START

M1
What is an algorithm?
- T/F 1–3 ›
- MC 1–2 ›
- OPEN none

M2
Sorting (bubble, selection, insertion)
- T/F 4–10 ›
- MC 3–6 ›
- OPEN 1–2 ›

M3
Searching and Big-O
- T/F 11–13 ›
- MC 7–10 ›
- OPEN 1, 3 ›

M4
BFS, DFS and greedy algorithms
- T/F 14–17 ›
- MC 11–15 ›
- OPEN 4–5 ›

M5
Arrays and linked lists
- T/F 18–19 ›
- MC 16–18 ›
- OPEN 6 ›

M6
Stacks, queues, recursion
- T/F 20–23 ›
- MC 19–23 ›
- OPEN 4 ›

M7
Trees and graphs
- T/F 24–27 ›
- MC 24–27 ›
- OPEN 7 ›

M8
Neurons and learning
- T/F 28–30 ›
- MC 28–30 ›
- OPEN 8 ›

M9
Activations and applications
- T/F 31–32 ›
- MC 31–33 ›
- OPEN 9 ›

M10
Forecasting
- T/F 33–35 ›
- MC 34–36 ›
- OPEN 10 ›

M11
Online learning and recommenders
- T/F 36–37 ›
- MC 37–39 ›
- OPEN 11 ›

FINAL BOSS

M12
Reinforcement learning and training
- T/F 38–40 ›
- MC 40–42 ›
- OPEN 12–13 ›



A B C ★ K

A

B

C

★

K

True or False

Forty quick-fire statements from every mission. Decide **true or false**, and fix the false ones in your own words.



MISSIONS COVERED · 12 OF 12

M1 M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 M12

QUESTIONS	FORMAT	MAX SCORE
40	T / F	40

BALANCE	AUTO-CHECK
20 / 20	ON

HOW TO PLAY

1 READ

Read each statement carefully. One wrong word makes it false.

2 TAP

Tap TRUE or FALSE. If false, write the fix on the line.

3 CHECK

Run the system check at the end to see your score.

PART A TRUE OR FALSE

Tap **TRUE** or **FALSE**. The right answer and its explanation appear **the moment you answer**. If false, say what would make it true in the fix-it line.

40 STATEMENTS · INSTANT ANSWERS · LIVE SCORE



A1 M1

An algorithm must finish after a finite number of steps.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A2 M1

An algorithm and a program are exactly the same thing.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A3 M1

Pseudocode must follow the exact syntax of one programming language.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A4 M2

After the first pass of bubble sort (ascending order), the largest value is in the last position.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A5 M2

Bubble sort with an early-exit check needs only one pass on a list that is already sorted.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A6 M2

In the worst case, bubble sort's running time grows as O(n log n).

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A7 M2

After the first pass of selection sort (ascending order), the smallest value is in the first position.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A8 M2

Selection sort makes fewer comparisons on an already-sorted list than on a list in reverse order.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A9 M2

Insertion sort on an already-sorted list of n items makes only n − 1 comparisons.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A10 M2

Insertion sort keeps a sorted part at the start of the list and inserts each next item into its correct place in that part.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A11 M3

Binary search works correctly on any list, whether it is sorted or not.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A12 M3

An O(1) operation takes about the same time no matter how many items there are.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A13 M3

If an algorithm is $O(n^2)$, doubling the input size roughly doubles the work.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A14 M4

BFS explores every cell one step away from the start before it explores any cell two steps away.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A15 M4

DFS always finds the route with the fewest steps.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A16 M4

Dijkstra's algorithm is an example of a greedy algorithm.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A17 M4

A greedy algorithm always finds the best possible solution.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A18 M5

Reading the item at index 2 of an array takes O(1) time.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A19 M5

In a singly linked list, you can jump directly to the 50th node without visiting the earlier nodes.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A20 M6

A stack follows the first in, first out (FIFO) rule.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A21 M6

BFS uses a queue to decide which location to explore next.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A22 M6

A recursive function can work correctly without a base case, as long as the input is small.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A23 M6

A deque (double-ended queue) allows items to be added and removed at both ends.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A24 M7

In a binary search tree, every value in a node's left subtree is smaller than that node's value.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A25 M7

In an undirected graph, an edge can only be followed in one direction.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A26 M7

Inserting the keys 1, 3, 8, 10 into an empty binary search tree in that order produces a balanced tree.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A27 M7

In a binary search tree, the successor of a key is the next larger key in sorted order.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A28 M8

In an artificial neuron, the bias is multiplied by one of the inputs before it is added.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A29 M8

A single neuron that draws one straight line can learn XOR.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A30 M8

A dense layer with 3 inputs and 4 neurons has 16 parameters.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A31 M9

Stacking many layers without non-linear activation functions still gives a network that can only learn linear (straight-line) relationships.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A32 M9

A defect detector that is 98% accurate on bottles of which 2% are defective must be good at finding defects.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A33 M10

Before testing a forecasting model, the time series should be shuffled randomly and then split into training and test sets.

TRUE FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

TEST IN PROGRESS

Systems check: all green!

More questions on the next page.

NEXT PAGE >>



A34 M10

Actual demand should be at or below a well-calibrated P90 forecast about 90% of the time.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A35 M10

A positive forecast bias means the forecasts are, on average, too low.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A36 M11

In online learning, the model is updated after each new example arrives.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A37 M11

Collaborative filtering needs a description of each item's content, such as its genre, to make recommendations.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A38 M12

In reinforcement learning, the agent is told the correct action for each state.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A39 M12

With a discount factor of γ = 0, the agent cares only about the immediate reward.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW

A40 M12

A training loss far lower than the validation loss is a sign of overfitting.

○ TRUE ○ FALSE IF FALSE, FIX IT: KEY ›

Tap TRUE or FALSE to unlock the answer SHOW



LIVE SCOREBOARD · PART A

Your score updates as you answer.

Each card marks itself ✓ or ✗ and opens its explanation the moment you answer.

Waiting for your first answer...

RESET PART A

SCORE ● LIVE

0 / 40

PART A COMPLETE

Level cleared!

Up next: Part B: multiple choice



GO TO PART B >>

A
B
C
★
K

Multiple Choice

Forty-two questions with four options each. Pick **one answer**, or tick **every** correct box on the select-all cards.



MISSIONS COVERED · 12 OF 12

M1 M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 M12

QUESTIONS	OPTIONS	MAX SCORE
42	A-D	42

SELECT-ALL	AUTO-CHECK
2	ON

HOW TO PLAY

1 THINK

Work it out on paper before you look at the options.

2 PICK

Choose one option, or tick all that apply on square-box cards.

3 CHECK

Run the system check, then read the key for every red X.

PART B MULTIPLE CHOICE

Choose **one answer** unless the question says **Select all that apply**. The correct option lights up green and the reasoning appears as soon as you answer.

42 QUESTIONS · INSTANT ANSWERS · LIVE SCORE



B1 M1

Which of the following is **not** a required property of an algorithm?

- A It produces an output.
- B Its steps are precise and unambiguous.
- C It is written in Python.
- D It finishes after a finite number of steps.

KEY ›

Pick an option to unlock the answer

SHOW

B2 M1

In a flowchart, which shape is used for a yes/no decision such as "Is score > highest?"

- A Oval
- B Rectangle
- C Diamond
- D Parallelogram

KEY ›

Pick an option to unlock the answer

SHOW

B3 M2

Bubble sort (ascending) is applied to [5, 1, 4]. What is the list after the first pass?

- A [1, 5, 4]
- B [1, 4, 5]
- C [5, 4, 1]
- D [4, 1, 5]

KEY ›

Pick an option to unlock the answer

SHOW

B4 M2

How many comparisons does bubble sort with early exit make on the already-sorted list [29, 40, 72, 85, 91, 100]?

- A 0
- B 5
- C 15
- D 36

KEY ›

Pick an option to unlock the answer

SHOW

B5 M2

Selection sort (ascending) is applied to [4, 3, 1, 2]. What is the list after the first pass?

- A [1, 3, 4, 2]
- B [3, 1, 2, 4]
- C [1, 2, 3, 4]
- D [1, 4, 3, 2]

KEY ›

Pick an option to unlock the answer

SHOW

B6 M2

Insertion sort is working on [40, 72, 85, 29]. The first three cards are already in order. How many comparisons does it make to insert 29?

- A 1
- B 3
- C 4
- D 6

KEY ›

Pick an option to unlock the answer

SHOW



B7 M3

Binary search looks for 14 in [2, 4, 6, 8, 10, 12, 14, 16], always checking the lower middle when there is an even number of items. Which values does it check, in order?

- A 2, 4, 6, 8, 10, 12, 14
- B 10, 14
- C 8, 14
- D 8, 12, 14

KEY ›

Pick an option to unlock the answer

SHOW

B8 M3

A sorted list has 1,024 items. How many times can binary search halve the range before only one item is left?

- A 10
- B 32
- C 512
- D 1,024

KEY ›

Pick an option to unlock the answer

SHOW

B9 M3

Which growth rate means the most work for very large n?

- A O(1)
- B O(log n)
- C O(n)
- D $O(n^2)$

KEY ›

Pick an option to unlock the answer

SHOW

TEST IN PROGRESS

You've got this!

More questions on the next page.

NEXT PAGE >>



B10 M3

Which operation takes O(n) time in the worst case?

- A Reading an array item by its index
- B Linear search in an unsorted list
- C Pushing onto a stack
- D Dequeuing the front item of a queue

KEY ›

Pick an option to unlock the answer

SHOW

B11 M4

During BFS on the B.O.T.A.K. maze, what does the number written in each cell mean?

- A The order in which DFS visited it
- B The number of walls next to it
- C Its row number
- D Its number of steps from S along a shortest route

KEY ›

Pick an option to unlock the answer

SHOW

B12 M4

A tree has root A; A's children are B and C; B's children are D and E; C's child is F. Visiting left neighbours first, which is a BFS order from A?

- A A, B, D, E, C, F
- B A, B, C, D, E, F
- C A, C, F, B, E, D
- D D, E, B, F, C, A

KEY ›

Pick an option to unlock the answer

SHOW

B13 M4

Using the same tree and visiting left neighbours first, which is a DFS order from A?

- A A, B, C, D, E, F
- B A, C, B, F, E, D
- C F, E, D, C, B, A
- D A, B, D, E, C, F

KEY ›

Pick an option to unlock the answer SHOW

B14 M4

Dijkstra's algorithm runs on a graph with edges S–A (1), A–G (8), S–B (2), B–C (2) and C–G (2). What is the cheapest cost from S to G?

- A 9
- B 4
- C 6
- D 5

KEY ›

Pick an option to unlock the answer SHOW

B15 M4

Coins come in 1, 3 and 4. Botak-Bot gives 6 in change using the greedy rule "take the biggest coin that still fits". How many coins does it use, and is that the fewest possible?

- A 2 coins; yes, it is the fewest
- B 3 coins; no, 2 coins is possible
- C 3 coins; yes, it is the fewest
- D 4 coins; no, 3 coins is possible

KEY ›

Pick an option to unlock the answer SHOW



B16 M5

A program reads items by their position very often and almost never inserts items in the middle. Which structure fits best?

- A Array
- B Singly linked list
- C Stack
- D Queue

KEY ›

SHOW

Pick an option to unlock the answer

B17 M5

Which statement about a singly linked list is correct?

- A Each node stores a value and a link to the next node.
- B Its nodes must be stored next to each other in memory.
- C Reaching the k-th node takes O(1) time.
- D It cannot grow after it is created.

KEY ›

SHOW

Pick an option to unlock the answer

B18 M5

The array [72, 85, 91] has room for one more item. What is the array after inserting 80 at index 1?

- A [80, 72, 85, 91]
- B [72, 85, 80, 91]
- C [72, 80, 91, 85]
- D [72, 80, 85, 91]

KEY ›

SHOW

Pick an option to unlock the answer

B19 M6

Start with an empty stack. Do: push A, push B, pop, push C, push D, pop. Which item is on top now?

- A A
- B B
- C C
- D D

KEY ›

Pick an option to unlock the answer SHOW

B20 M6

Start with an empty queue. Do: enqueue Ana, enqueue Ben, enqueue Cai, dequeue, enqueue Dee, dequeue. Who is at the front now?

- A Ana
- B Ben
- C Cai
- D Dee

KEY ›

Pick an option to unlock the answer SHOW

B21 M6

Select all that apply. Which are typical uses of a stack?

- [ ] A Undo history in an editor
- [ ] B Checking that brackets match
- [ ] C Serving print jobs in the order they arrive
- [ ] D Keeping track of function calls

SELECT ALL THAT APPLY

KEY ›

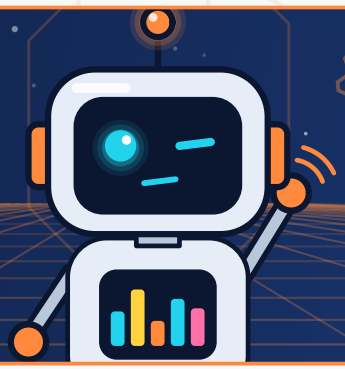
Tick all the correct boxes to unlock the answer SHOW

TEST IN PROGRESS

Systems check: all green!

More questions on the next page.

NEXT PAGE >>



B22 M6

factorial(n) returns 1 when n == 1, and otherwise returns n × factorial(n − 1). When factorial(3) runs, which call finishes (returns) first?

- A factorial(3)
- B factorial(2)
- C factorial(1)
- D They all finish at the same time

KEY ›

SHOW

Pick an option to unlock the answer

B23 M6

Which describes an input-restricted deque?

- A Items are added at one end only, but can be removed from both ends.
- B Items can be added at both ends, but are removed from one end only.
- C Items are removed in order of priority.
- D Items can only be added when the deque is empty.

KEY ›

SHOW

Pick an option to unlock the answer

B24 M7

In a friendship network, Nicky is linked to Darren and Neil, Neil is linked to Glen, and Darren is linked to Glen. Using BFS from Nicky, how many edges are on the shortest route to Glen?

- A 1
- B 3
- C 2
- D 4

KEY ›

SHOW

Pick an option to unlock the answer

B25 M7

A binary search tree has root 8, left child 3 (whose left child is 1) and right child 10. Where is 5 inserted?

- A Right child of 3
- B Left child of 1
- C Left child of 10
- D It becomes the new root

KEY ›

Pick an option to unlock the answer SHOW

B26 M7

A BST has root 8 with children 3 and 10; 3 has left child 1 and right child 6, and 6 has right child 7. If 3 is deleted using its successor, which key takes its place?

- A 1
- B 6
- C 7
- D 8

KEY ›

Pick an option to unlock the answer SHOW

B27 M7

In the adjacency matrix of the friendship network (Nicky–Darren, Nicky–Neil, Darren–Glen, Neil–Glen), what is in the cell for row Nicky, column Glen?

- A 1
- B 2
- C 0
- D Darren, Neil

KEY ›

Pick an option to unlock the answer SHOW

TEST IN PROGRESS

Botak-Bot believes in you!



B28 M8

A ReLU neuron has inputs [1, 3], weights [0.5, −0.5] and bias 0.2. What is its output?

- A 0
- B −0.8
- C 0.8
- D 2.2

KEY ›

Pick an option to unlock the answer

SHOW

B29 M8

How many parameters (weights and biases) does a dense layer with 4 inputs and 2 neurons have?

- A 8
- B 6
- C 12
- D 10

KEY ›

Pick an option to unlock the answer

SHOW

B30 M8

A weight is 0.5, its gradient is 2 and the learning rate is 0.1. What is the weight after one gradient-descent step?

- A 0.7
- B 0.3
- C 0.48
- D 0.2

KEY ›

Pick an option to unlock the answer

SHOW

TEST IN PROGRESS

Halfway to Commander!



B31 M9

A network must decide which of 10 digits (0–9) is in an image. Which output activation fits best?

- A Sigmoid
- B Softmax
- C Linear (no activation)
- D Softplus

KEY ›

SHOW

Pick an option to unlock the answer

B32 M9

Why is ReLU usually preferred to sigmoid in the hidden layers of a deep network?

- A ReLU always outputs values between 0 and 1.
- B ReLU makes the network linear, which is easier to train.
- C ReLU's slope is 1 for positive inputs, so the learning signal shrinks less across many layers.
- D ReLU never outputs 0.

KEY ›

SHOW

Pick an option to unlock the answer

B33 M9

Select all that apply. Which situations suit a neural network well?

- [ ] A Recognising objects in thousands of labelled photos
- [ ] B Converting speech to text
- [ ] C Adding up a short list of prices exactly
- [ ] D Predicting from 10 rows of data when every decision must be fully explained

SELECT ALL THAT APPLY

KEY ›

SHOW

Tick all the correct boxes to unlock the answer

B34 M10

Actual sales are [10, 20, 30] and the forecasts are [12, 18, 33]. What is the mean absolute error (MAE)?

- A 1
- B 7
- C 2.33
- D 0.33

KEY ›

Pick an option to unlock the answer

SHOW

B35 M10

Running out of a cake loses 4 in profit, and each leftover cake wastes 1. Using the newsvendor rule, which quantile of the demand forecast should the shop stock to?

- A P80
- B P20
- C P50
- D P100

KEY ›

Pick an option to unlock the answer

SHOW

B36 M10

A P90 ($\tau = 0.9$) forecast is 110 and the actual value is 100. What is the pinball loss?

- A 9
- B 10
- C 0
- D 1

KEY ›

Pick an option to unlock the answer

SHOW



B37 M11

The mean of 5 values is 6. A sixth value, 12, arrives. What is the new mean, updated without storing the old values?

- A 9
- B 7
- C 6.5
- D 12

KEY ›

Pick an option to unlock the answer

SHOW

B38 M11

A recommender shows 5 films. 2 of them are relevant to the user, who has 4 relevant films in total. What are Precision@5 and Recall@5?

- A 0.5 and 0.4
- B 0.4 and 0.4
- C 0.5 and 0.5
- D 0.4 and 0.5

KEY ›

Pick an option to unlock the answer

SHOW

B39 M11

Two users rate three films: A = [3, 4, 0] and B = [4, 3, 0]. What is their cosine similarity?

- A 0.96
- B 1
- C 0.5
- D 0

KEY ›

Pick an option to unlock the answer

SHOW

TEST IN PROGRESS

You've got this!



B40 M12

Q-learning: Q(s, a) = 0.5, α = 0.2, γ = 0.9, reward r = 1, and the best Q-value in the next state is 1. What is the new Q(s, a)?

- A 1.9
- B 0.38
- C 1.4
- D 0.78

KEY ›

SHOW

Pick an option to unlock the answer

B41 M12

An agent uses the ε-greedy rule with ε = 0.1. What does it do?

- A It always takes the action with the highest Q-value.
- B It takes a random action about 90% of the time.
- C It stops training when its reward falls below 0.1.
- D It takes a random action about 10% of the time and the best-known action otherwise.

KEY ›

SHOW

Pick an option to unlock the answer

B42 M12

After training, the training loss and the validation loss are both high and close together. What is the most likely diagnosis?

- A Underfitting
- B Overfitting
- C A perfect fit
- D The test set was used for tuning

KEY ›

SHOW

Pick an option to unlock the answer



LIVE SCOREBOARD · PART B

Your score updates as you answer.

Each card marks itself ✓ or ✗ and opens its explanation the moment you answer.

Waiting for your first answer...

RESET PART B

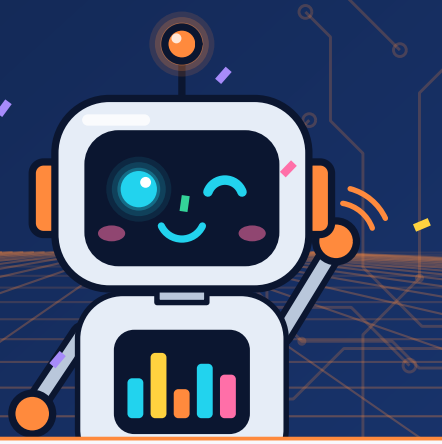
SCORE · LIVE

0 / 42

PART B COMPLETE

Level cleared!

Up next: Part C: open-ended



GO TO PART C >>

Open-Ended

Thirteen longer questions. Explain your reasoning, show your working and aim for **100 words or more** on each.



MISSIONS COVERED · 11 OF 12

M1 M2 M3 M4 M5 M6 M7 M8 M9 M10 M11 M12

QUESTIONS	TARGET	WORD COUNT
13	100+ W	LIVE

MARKING	PARTS
KEY	(a)-(d)

HOW TO PLAY

1 PLAN

List the key points and numbers you need before writing.

2 WRITE

Type your answer. The counter shows when you pass 100 words.

3 COMPARE

Compare with the key points and marking notes in the key.

PART C OPEN-ENDED

Answer in full sentences. Aim for at least 100 words per question, or equivalent working (tables, diagrams, calculations). Show your reasoning: the explanation earns more credit than the final answer alone.

13 QUESTIONS | AIM: 100+ WORDS EACH | MODEL ANSWERS



C1

M2

M3

Bubble sort with early exit is run on two lists: [72, 85, 40, 29, 100, 91] and [29, 40, 72, 85, 91, 100]. Trace both runs (passes, comparisons and swaps). Use your results to explain the difference between bubble sort's best case and worst case, and why it is usually described as $O(n^2)$.

WORD COUNTER 0 / 100 words

KEY ›

C1 · MODEL ANSWER

Write your answer first, then reveal the model answer

SHOW MODEL ANSWER

C2

M2

Selection sort and insertion sort are both run on [72, 85, 40, 29, 100, 91]. Trace both (the list after each pass or step, with comparisons and swaps or shifts). Then a class leaderboard is already sorted and one new score is added at the end: explain which of the two sorts you would use to put it in place, and why.

WORD COUNTER 0 / 100 words

KEY ›

C2 · MODEL ANSWER

Write your answer first, then reveal the model answer

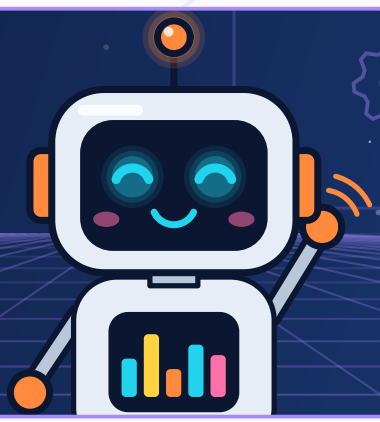
SHOW MODEL ANSWER

TEST IN PROGRESS

You've got this!

More questions on the next page.

NEXT PAGE >>



C3

M3

A school database has 1,000,000 student IDs. Explain why binary search needs at most about 20 checks while linear search may need 1,000,000. Then describe one situation in which you would still choose linear search, and justify your choice.

WORD COUNTER 0 / 100 words

KEY ›

C3 · MODEL ANSWER

Write your answer first, then reveal the model answer

SHOW MODEL ANSWER

TEST IN PROGRESS

Botak-Bot believes in you!

More questions on the next page.

NEXT PAGE >>



C4

M4

M6

BFS stores waiting locations in a queue. Explain how the queue makes BFS explore "in waves" of increasing distance. Then explain what would change if the queue were replaced by a stack, using the A–F tree (A → B, C; B → D, E; C → F) or the B.O.T.A.K. maze as an example.

WORD COUNTER 0 / 100 words

KEY ›

C4 · MODEL ANSWER

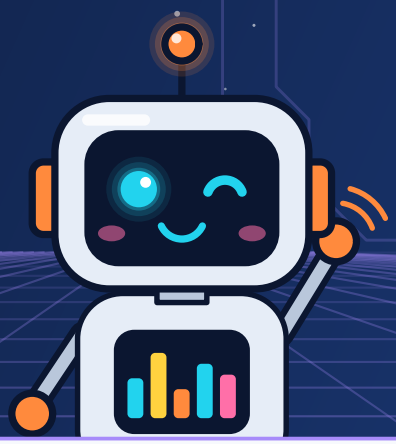
Write your answer first, then reveal the model answer

SHOW MODEL ANSWER

TEST IN PROGRESS

Halfway to Commander!

More questions on the next page.



NEXT PAGE >>

C5

M4

Use the graph with edges S–A (1), A–G (8), S–B (2), B–C (2) and C–G (2).

(a) Which route does BFS find from S to G, and what does it cost?

(b) Trace Dijkstra's algorithm round by round and give the cheapest route.

(c) A "greedy walker" always takes the cheapest edge out of its current vertex: what does it do?

(d) Both Dijkstra's algorithm and the walker are greedy. Explain why only one of them is guaranteed to find the cheapest route.

WORD COUNTER 0 / 100 words

KEY ›

C5 · MODEL ANSWER

Write your answer first, then reveal the model answer

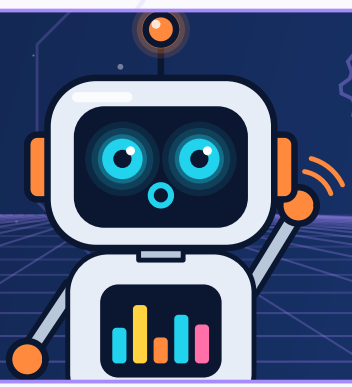
SHOW MODEL ANSWER

TEST IN PROGRESS

Keep going, pilot!

More questions on the next page.

NEXT PAGE >>



C6

M5

A music app has two features: (1) a leaderboard where the app constantly reads "the song at position k"; (2) a playlist editor where users often insert songs in the middle of the current list. Choose a data structure for each feature and justify each choice using the cost of the important operations. Mention one disadvantage of each choice.

WORD COUNTER 0 / 100 words

KEY ›

C6 · MODEL ANSWER

Write your answer first, then reveal the model answer

SHOW MODEL ANSWER

TEST IN PROGRESS

Systems check: all green!

More questions on the next page.

NEXT PAGE >>



C7

M7

Start with the binary search tree with root 8, left child 3 (whose left child is 1) and right child 10. Insert 6, then 9, then 14, explaining each decision. Draw the final tree and write its inorder traversal. Then explain why searching a BST is usually faster than checking every value, and what goes wrong if the same seven values are inserted in increasing order.

WORD COUNTER 0 / 100 words

KEY ›

C7 · MODEL ANSWER

Write your answer first, then reveal the model answer

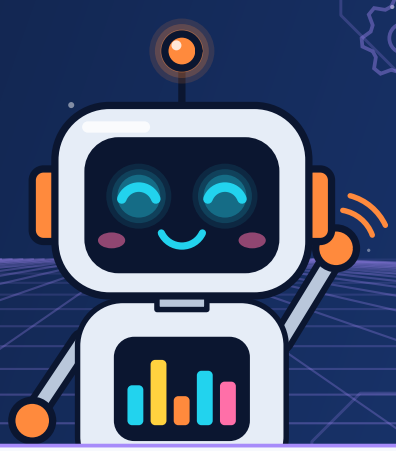
SHOW MODEL ANSWER

TEST IN PROGRESS

You've got this!

More questions on the next page.

NEXT PAGE >>



C8

M8

A neuron has no activation: $\hat{y} = w \times x + b$, with $w = 0.5$, $b = 0$ and squared-error loss $L = (\hat{y} - y)^2$. One training example has $x = 2$ and target $y = 3$.

(a) Do the forward pass and calculate the loss.

(b) The gradient for w is $2(\hat{y} - y) \times x$. Calculate it and update w with learning rate 0.1.

(c) Calculate the new prediction and loss.

(d) Explain in your own words what each step of training (forward pass, loss, backward pass, update) does, and why many such steps over many examples are needed.

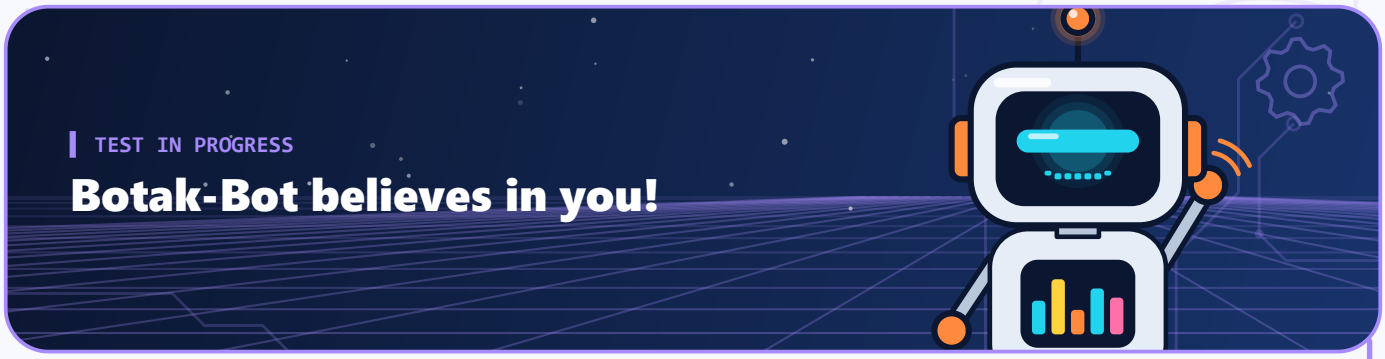
WORD COUNTER 0 / 100 words

KEY ›

C8 · MODEL ANSWER

Write your answer first, then reveal the model answer

SHOW MODEL ANSWER



C9

M9

Botak-Bot builds a network with 10 hidden layers that have no activation functions, followed by one sigmoid output neuron.

(a) Explain, using algebra or a clear example, why the 10 hidden layers add no extra power: the network can still only draw a straight-line boundary.

(b) After adding sigmoid to every hidden layer, the early layers barely learn. Explain why, using sigmoid's maximum slope of 0.25.

(c) Suggest two fixes.

(d) Choose an output activation for each: a house price; spam or not spam; one of 5 animal species; the standard deviation of a forecast.

WORD COUNTER 0 / 100 words

KEY ›

C9 · MODEL ANSWER

Write your answer first, then reveal the model answer

SHOW MODEL ANSWER

TEST IN PROGRESS

Halfway to Commander!

More questions on the next page.

NEXT PAGE >>



C10

M10

A bakery sells croissants. Week 1 (Mon–Fri): 30, 25, 28, 27, 40. Week 2: 32, 26, 29, 28, 42. Week 3 (actual): 31, 27, 31, 29, 45.

(a) Make a seasonal naive forecast for week 3 and calculate the errors (forecast − actual), MAE, WAPE and bias.

(b) Explain why a seasonal naive forecast is a sensible baseline for this data, and why the data must not be shuffled before testing a model.

(c) A croissant costs 1 to make and sells for 3. Which quantile of the forecast should the bakery stock to, and why is this above the median?

WORD COUNTER 0 / 100 words

KEY ›

C10 · MODEL ANSWER

Write your answer first, then reveal the model answer

SHOW MODEL ANSWER

TEST IN PROGRESS

Keep going, pilot!

More questions on the next page.

NEXT PAGE >>

C11

M11

A video app recommends videos to its users.
(a) Users A = [3, 4, 0], B = [4, 3, 0] and C = [0, 0, 5] rated three videos. Calculate the cosine similarity of A with B and of A with C, and explain which user's ratings should influence A's recommendations most.
(b) A new user joins with no history. Name this problem and describe two ways to handle it.
(c) Over a holiday, viewing habits change suddenly and the model's errors rise. Name this effect and explain how online learning and a sliding window of recent data help.
(d) Explain one risk of a feedback loop in recommendations.

WORD COUNTER 0 / 100 words

KEY ›

C11 · MODEL ANSWER

Write your answer first, then reveal the model answer

SHOW MODEL ANSWER

TEST IN PROGRESS

Systems check: all green!



C12 M12

A robot learns in a corridor of cells 0, 1, 2, 3; the goal is cell 3 (reward +1 on arriving, 0 for other moves). Q(s, right) starts at 0 everywhere, and Q at the goal is always 0. Use $\alpha = 1$ and $\gamma = 0.8$, and let the robot always move right from cell 0.

(a) Calculate Q(0, right), Q(1, right) and Q(2, right) after each of episodes 1, 2 and 3.

(b) Explain why value spreads backwards one cell per episode.

(c) Explain why a real agent also needs exploration, for example ε-greedy, and why the reward must be designed carefully.

WORD COUNTER 0 / 100 words

KEY ›

C12 · MODEL ANSWER

Write your answer first, then reveal the model answer

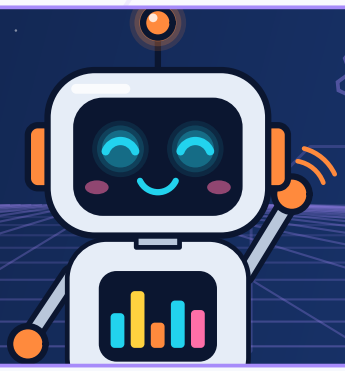
SHOW MODEL ANSWER

TEST IN PROGRESS

You've got this!

More questions on the next page.

NEXT PAGE >>



C13

M12

Four students train classifiers and report: (1) training loss 0.02, validation loss 0.45; (2) training and validation loss both stuck at 0.9; (3) the loss becomes NaN after a few steps; (4) a classifier for 10 equally common classes starts with a loss of 9.7. Diagnose each case and suggest a fix. Then explain the different jobs of the training, validation and test sets, and why tuning on the test set is a mistake.

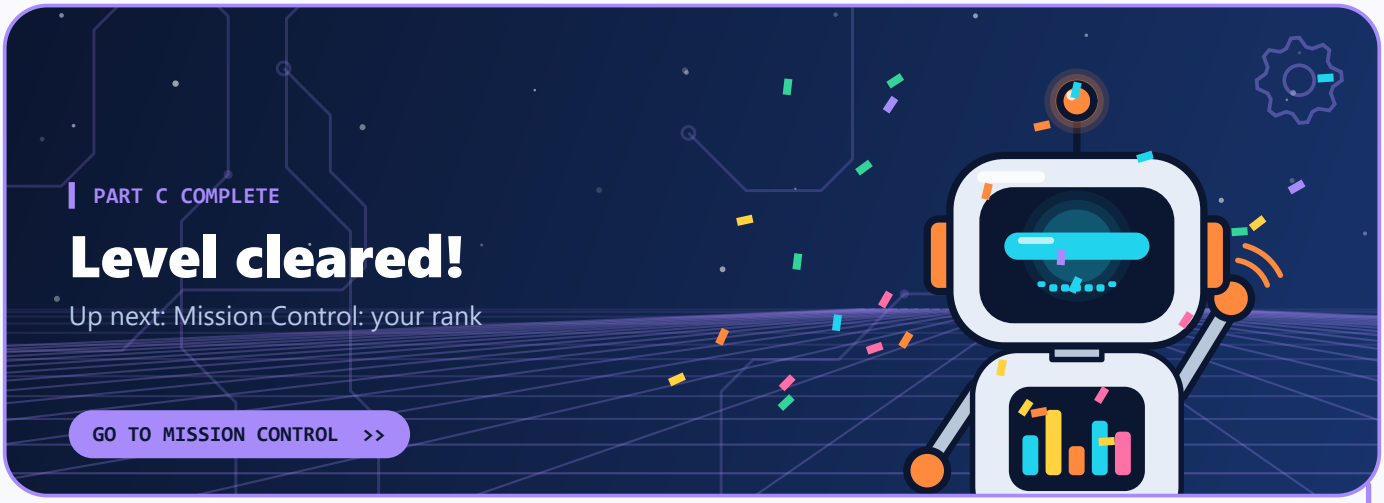
WORD COUNTER 0 / 100 words

KEY ›

C13 · MODEL ANSWER

Write your answer first, then reveal the model answer

SHOW MODEL ANSWER



MISSION CONTROL

How did your bot do?

Finished Parts A, B and C? Press **CALCULATE MY RANK**. Parts A and B are scored automatically; Part C counts how many answers reached 100 words. Mark Part C yourself with the key points in the answer key.



A
B
C
★
K

PART A	PART B	PART C
-- / 40	-- / 42	-- / 13
true or false	multiple choice	answers with 100+ words

YOUR RANK (PARTS A + B)

Awaiting launch...

Press the button to calculate.

CALCULATE MY RANK

RANK LADDER



ROOKIE BOT
0–39%



CADET BOT
40–69%



PILOT BOT
70–89%



COMMANDER BOT
90–100%

OPEN THE ANSWER KEY ›



B.O.T.A.K. · TEST BANK V1.0

Certificate

OF PILOT RANK

This certifies that pilot

fills in from the PILOT NAME box on the cover

completed the B.O.T.A.K. Test Bank and earned the rank of

PART A	PART B	PART C

ROOKIE	CADET	PILOT	COMMANDER
0-39%	40-69%	70-89%	90%+

DATE

BOTAK-BOT, MISSION CONTROL

PRINT CERTIFICATE

Answer Key

Every answer with its **explanation**. Read the reasoning even when you were right, then revisit the matching Study Guide mission.



PART A	PART B	PART C
40	42	13

TOTAL	BACK LINKS
95	ALL

HOW TO USE THE KEY

1 FIND

Jump to any answer with the KEY button on its question card.

2 LEARN

Read why the answer is right and why the others are not.

3 RETRY

Tap BACK, fix your answer, and run the check again.

ANSWER KEY CHECK YOUR WORK



Every answer is explained, so read the explanation even when you were right. Tap ‹ **BACK** on any row to return to its question.

PART A | PART B | PART C KEY POINTS | REVIEW RECORD

PART A · TRUE OR FALSE

Balance check: 20 true, 20 false.

A1	TRUE	Finiteness is a required property. Instructions that could run forever are not an algorithm.	‹ BACK
A2	FALSE	An algorithm is the step-by-step method. A program is one implementation of that method in a programming language; one algorithm can be written as many programs.	‹ BACK
A3	FALSE	Pseudocode is programming-like but has no fixed syntax. It describes the steps without the rules of a particular language.	‹ BACK
A4	TRUE	Each comparison carries the larger value to the right, so the largest value "bubbles" to the end in pass 1.	‹ BACK
A5	TRUE	The first pass makes no swaps, so the swapped flag stays false and the algorithm stops (n − 1 comparisons).	‹ BACK
A6	FALSE	Its worst case is $O(n^2)$: up to n − 1 passes, each comparing up to n − 1 pairs.	‹ BACK
A7	TRUE	Pass 1 scans the whole list for the smallest value and swaps it into the first position.	‹ BACK
A8	FALSE	Selection sort always scans the whole unsorted part to find the smallest value, so it makes n(n − 1)/2 comparisons whatever the order: 15 for six items, sorted or not.	‹ BACK
A9	TRUE	Each card after the first is compared once with its left neighbour, which is not bigger, so it stays where it is: n − 1 comparisons, O(n).	‹ BACK
A10	TRUE	That is how it works: like sorting cards in your hand, each new card moves left past the bigger cards and drops into the gap.	‹ BACK
A11	FALSE	Binary search needs sorted data. Discarding half the list is only safe because the order tells us which half cannot contain the target.	‹ BACK

A12 TRUE — O(1) means constant time: the work does not grow with n (e.g. reading an array item by its index). ‹ BACK

A13 FALSE — $(2n)^2 = 4n^2$, so the work roughly quadruples. It is O(n) algorithms whose work roughly doubles. ‹ BACK

A14 TRUE — BFS uses a queue, so it explores in order of distance: all of distance 1, then all of distance 2, and so on. ‹ BACK

A15 FALSE — DFS follows one branch as deep as it can. The first route it finds may be long; BFS is the one that guarantees the fewest steps when every move costs the same. ‹ BACK

A16 TRUE — Each round it settles the unsettled vertex with the smallest known total cost and never changes that choice. With no negative costs, this greedy choice is always safe. ‹ BACK

A17 FALSE — Greedy algorithms take the best-looking step each time, which can fail. With coins 1, 3 and 4, greedy change for 6 is 4 + 1 + 1 (3 coins), but 3 + 3 uses only 2. ‹ BACK

A18 TRUE — Array items are stored side by side, so the computer calculates the position and jumps straight to it. ‹ BACK

A19 FALSE — Nodes are found by following links from the head, so reaching the 50th node means passing through the 49 before it (O(n)). ‹ BACK

A20 FALSE — A stack is last in, first out (LIFO): the most recently pushed item is popped first. A queue is FIFO. ‹ BACK

A21 TRUE — The queue makes earlier-discovered (closer) locations come out first, which gives BFS its level-by-level order. ‹ BACK

A22 FALSE — Without a base case the function never stops calling itself, whatever the input; the call stack grows until the program fails (in Python, a RecursionError). Every recursive function needs a base case. ‹ BACK

A23 TRUE — That is what "double-ended" means. Input-restricted and output-restricted deques limit one of the two operations to one end. ‹ BACK

A24 TRUE — That is the BST rule: left subtree smaller, right subtree larger. It lets a search go left or right at each node. ‹ BACK

A25 FALSE — Undirected edges work both ways, like a friendship: if Nicky is linked to Neil, Neil is linked to Nicky. Edges with one direction belong to a directed graph. ‹ BACK

A26 FALSE — Each new key is larger than all the others, so each becomes a right child: the tree is a chain of height 4, and searching it is as slow as a linked list (O(n)). ‹ BACK

A27 TRUE — If the node has a right subtree, the successor is the smallest key there; otherwise it is the first ancestor reached from a left child. ‹ BACK

A28 FALSE — Each input is multiplied by its weight; the bias is then added on its own. It shifts the result, like the b in y = mx + b. ‹ BACK

A29 FALSE — XOR's two 1s are on opposite corners, (0, 1) and (1, 0), and no single straight line separates them from (0, 0) and (1, 1). A hidden layer is needed. ‹ BACK

A30 TRUE — 3 × 4 = 12 weights plus 4 biases = 16. ‹ BACK

A31 TRUE — Each layer computes Wx + b, and two such layers combine into one: $W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$. Any number of them is still one linear function. ‹ BACK

A32 FALSE — A model that always answers "ok" is 98% accurate on this data and finds no defects at all. For rare events, check recall: the share of real defects that are found. ‹ BACK

A33 FALSE — Time series must be split by time: train on the past and test on the future. Shuffling lets the model learn from days that come after the test days, so the results look better than real use. ‹ BACK

A34 TRUE — That is the meaning of the 90% quantile. If demand is below it much less often, the forecast is too low; much more often, too high. ‹ BACK

A35 FALSE — Bias = average of (forecast − actual). A positive bias means the forecasts are too high on average; a negative bias means too low. ‹ BACK

A36 TRUE — Online learning updates after every single example and then discards it. Batch learning trains on all the data at once. ‹ BACK

A37 FALSE — Collaborative filtering uses only the pattern of ratings or clicks across users. Content-based filtering is the method that uses item features. ‹ BACK

A38 FALSE — The agent only receives rewards, often only at the end. It must discover good actions by trial and error. Being told the correct answer is supervised learning. ‹ BACK

A39 TRUE — Later rewards are multiplied by γ, γ^2, ..., which are all 0. A γ close to 1 makes the agent far-sighted. ‹ BACK

A40 TRUE — The model fits the training data much better than new data, so it has memorised details instead of learning the general pattern. ‹ BACK

PART B · MULTIPLE CHOICE

B1	C	An algorithm can be written in any form (steps, pseudocode, flowchart) or any language. A, B and D are required properties: output, definiteness and finiteness.	‹ BACK
B2	C	Diamond = decision. Oval = start/end, rectangle = process step, parallelogram = input/output.	‹ BACK
B3	B	5 > 1, swap → [1, 5, 4]; 5 > 4, swap → [1, 4, 5]. Option A is the list after only the first comparison.	‹ BACK
B4	B	One pass compares the 5 neighbouring pairs, makes no swaps, and stops. Without the early-exit check it would always make all 5 passes: 5 + 4 + 3 + 2 + 1 = 15 comparisons, even on a sorted list.	‹ BACK
B5	A	Pass 1 finds the smallest value, 1, and swaps it with the first card, 4: [1, 3, 4, 2]. Option B is one pass of bubble sort, option C is the fully sorted list, and option D shifts the others instead of swapping.	‹ BACK
B6	B	29 is compared with 85, 72 and 40. Each is bigger, so each shifts right and 29 reaches the front: 3 comparisons, 3 shifts.	‹ BACK
B7	D	Middle of 8 items = 8 (14 is larger, keep 10–16); middle = 12 (keep 14–16); middle = 14, found. Option A is linear search.	‹ BACK
B8	A	1,024 → 512 → 256 → 128 → 64 → 32 → 16 → 8 → 4 → 2 → 1 is 10 halvings, because 2^{10} = 1,024.	‹ BACK
B9	D	For large n: 1 < log n < n < n^2. For n = 1,000,000, n^2 is a trillion.	‹ BACK
B10	B	Linear search may check all n items. The other three are O(1) operations.	‹ BACK
B11	D	BFS numbers each cell with its distance from S, wave by wave. That is why G's number (7) is the length of the shortest route.	‹ BACK
B12	B	BFS goes level by level: A (distance 0), B and C (distance 1), then D, E, F (distance 2). Option A is the DFS order.	‹ BACK
B13	D	DFS goes as deep as possible first: A → B → D, back to B → E, back to A → C → F.	‹ BACK
B14	C	Settle S (0) → A = 1, B = 2; settle A → G = 9; settle B → C = 4; settle C → G = min(9, 6) = 6. The route is S → B → C → G. Option A (9) is the fewest-edges route S → A → G that BFS would choose.	‹ BACK
B15	B	Greedy takes 4, then 1, then 1: 3 coins. But 3 + 3 = 6 uses only 2 coins, so the greedy choice of 4 was not part of the best answer.	‹ BACK

B16	A	Arrays give O(1) access by index. Linked lists need O(n) to reach a position, and stacks and queues only give access at their ends.	‹ BACK
B17	A	That is the definition of a node. Nodes can be anywhere in memory (B false), reaching the k-th node takes O(n) (C false), and lists grow easily by adding nodes (D false).	‹ BACK
B18	D	Index 1 is the second position. 85 and 91 each shift one place right to make room, keeping their order.	‹ BACK
B19	C	[A] → [A, B] → pop B → [A] → [A, C] → [A, C, D] → pop D → [A, C]. The top is C.	‹ BACK
B20	C	Ana, Ben, Cai → Ana leaves → Ben, Cai, Dee → Ben leaves → Cai, Dee. Cai is at the front.	‹ BACK
B21	A, B, D	Undo, bracket matching and function calls all need the most recent item first (LIFO). Print jobs in arrival order need a queue (FIFO).	‹ BACK
B22	C	factorial(3) waits for factorial(2), which waits for factorial(1). The base case factorial(1) is the last call pushed onto the call stack and the first to return (LIFO); then factorial(2) returns 2, then factorial(3) returns 6.	‹ BACK
B23	A	Input (adding) is restricted to one end; removing is allowed at both ends. Option B describes an output-restricted deque, and option C a priority queue.	‹ BACK
B24	C	Distance 1: Darren and Neil. Distance 2: Glen (Nicky → Neil → Glen or Nicky → Darren → Glen). Nicky and Glen are not linked directly, so 1 is wrong.	‹ BACK
B25	A	5 < 8, go left to 3; 5 > 3, go right; that spot is empty, so 5 becomes the right child of 3.	‹ BACK
B26	B	3 has two children, so it is replaced by its successor, the smallest key in its right subtree: 6. Then 7 moves up into the old place of 6. Inorder afterwards: 1, 6, 7, 8, 10.	‹ BACK
B27	C	0, because there is no edge between Nicky and Glen. They are two edges apart, but a matrix cell only records a direct edge. Option D is what an adjacency list stores for Nicky.	‹ BACK
B28	A	z = 0.5 × 1 − 0.5 × 3 + 0.2 = −0.8. ReLU turns negative values into 0, so the output is 0. Option B forgets the activation.	‹ BACK
B29	D	4 × 2 = 8 weights plus 2 biases = 10. Option A forgets the biases.	‹ BACK
B30	B	new w = w − learning rate × gradient = 0.5 − 0.1 × 2 = 0.3. A positive gradient means increasing w would raise the loss, so w goes down. Option A moves in the wrong direction.	‹ BACK

B31	B	Softmax turns the 10 scores into probabilities that add up to 1, for choosing one class out of many. Sigmoid suits one yes/no output, linear suits any number, and softplus suits a number that must be above 0.	‹ BACK
B32	C	Sigmoid's slope is at most 0.25, so multiplying it across many layers makes the gradient vanish. ReLU is not bounded above (A false), is non-linear (B false), and outputs 0 for negative inputs (D false).	‹ BACK
B33	A, B	Neural networks shine with lots of data and messy inputs such as images and audio. Exact arithmetic needs a simple rule, and a tiny dataset that must be fully explained suits a simpler model.	‹ BACK
B34	C	The errors (forecast − actual) are +2, −2 and +3; their absolute values add up to 7, and 7 / 3 ≈ 2.33. Option A is the bias (average signed error), and option B is the total before dividing.	‹ BACK
B35	A	Stock to shortage cost ÷ (shortage cost + leftover cost) = 4 / (4 + 1) = 0.8, the P80. Running out costs four times more than a leftover, so it pays to stock above the median (P50).	‹ BACK
B36	D	The forecast is too high by 10, so the loss is (1 − 0.9) × 10 = 1. If it had been too low by 10, the loss would be 0.9 × 10 = 9 (option A).	‹ BACK
B37	B	new mean = old mean + (new value − old mean) / n = 6 + (12 − 6) / 6 = 7. Check: (5 × 6 + 12) / 6 = 42 / 6 = 7. Option A averages only the old mean and the new value.	‹ BACK
B38	D	Precision@5 = relevant shown / shown = 2 / 5 = 0.4. Recall@5 = relevant shown / all relevant = 2 / 4 = 0.5. Option A swaps them.	‹ BACK
B39	A	$A \cdot B = 12 + 12 + 0 = 24$; $\lvert A\rvert = \lvert B\rvert = \sqrt{25} = 5$; 24 / (5 × 5) = 0.96. Their tastes are very similar but not identical, so the answer is close to but below 1.	‹ BACK
B40	D	target = r + γ × max Q(s′, a′) = 1 + 0.9 × 1 = 1.9. new Q = 0.5 + 0.2 × (1.9 − 0.5) = 0.5 + 0.28 = 0.78. Option A is the target itself, which the update only moves part of the way towards.	‹ BACK
B41	D	With probability ε = 0.1 it explores (acts randomly); otherwise it exploits the best-known action. Option A would never explore, and option B swaps exploring and exploiting.	‹ BACK
B42	A	High error on both sets means the model has not learned the pattern: it is underfitting. Try a bigger model, train longer or improve the features. Overfitting shows low training error but high validation error.	‹ BACK

PART C · OPEN-ENDED

Expected key points

C1

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

M2 M3

- Unsorted list: pass 1 → [72, 40, 29, 85, 91, 100] (5 comparisons, 3 swaps); pass 2 → [40, 29, 72, 85, 91, 100] (4, 2); pass 3 → sorted (3, 1); pass 4 makes no swaps (2, 0) and stops. Total 14 comparisons, 6 swaps.
- Sorted list: 1 pass, 5 comparisons, 0 swaps, then early exit.
- Best case (already sorted) is O(n) because of the early exit. Worst case (reverse order) needs $n(n-1)/2$ comparisons, which grows as $O(n^2)$; average cases also grow as $O(n^2)$.
- Big-O describes growth for large inputs, so bubble sort is usually summarised by its worst/average case, $O(n^2)$.

Marking note: Accept a correct trace in any clear format (table or list). Award full credit only if the counts are right and the answer links them to best vs worst case.

C2

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

M2

- Selection sort: passes select 29, 40, 72, 85 (already in place, no swap), 91 → [29, 85, 40, 72, 100, 91], [29, 40, 85, 72, 100, 91], [29, 40, 72, 85, 100, 91], unchanged, [29, 40, 72, 85, 91, 100]. 5 + 4 + 3 + 2 + 1 = 15 comparisons, 4 swaps.
- Insertion sort: insert 85 (1 comparison, 0 shifts), 40 (2, 2), 29 (3, 3), 100 (1, 0), 91 (2, 1). Total 9 comparisons, 6 shifts.
- Selection sort always scans the whole unsorted part, so it makes $n(n-1)/2$ comparisons on any list. Insertion sort stops moving a card when it meets one that is not bigger: $n-1$ comparisons on a sorted list ($O(n)$), $n(n-1)/2$ in reverse order ($O(n^2)$).
- Leaderboard: insertion sort. The old scores are already in order, so each one after the first needs only one comparison, and the new score moves left only past the bigger scores. Selection sort would still scan everything on every pass.

Marking note: Accept any clear trace format. Full credit needs correct counts for both sorts and a choice justified by how the starting order affects each sort.

DEBRIEF IN PROGRESS

Botak-Bot believes in you!

More questions on the next page.

NEXT PAGE >>



C3

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

M3

- Binary search halves the remaining range at every check; 2^{20} = 1,048,576 ≥ 1,000,000, so about 20 halvings leave one item. This is O(log n).
- Linear search checks items one by one; if the ID is last or missing it checks all 1,000,000 (O(n)).
- Binary search requires sorted data. Valid reasons to choose linear search: the data is unsorted and will only be searched once (sorting first would cost more than one linear search); the list is very small; items are added constantly in no order.

★ *Strong answers mention that the speed advantage grows as n grows.*

Marking note: Any reasonable situation is acceptable if it is justified by sortedness, list size, or how often the list is searched.

C4

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

M4

M6

- Queue = FIFO: locations are explored in the order they were discovered. All distance-1 locations are discovered before any distance-2 locations, so they come out of the queue first; this creates the waves.
- With a stack (LIFO), the most recently discovered location is explored next, so the search dives down one branch before returning. This is depth-first search.
- Tree example: queue gives A, B, C, D, E, F; stack-based DFS (left first) gives A, B, D, E, C, F.
- Consequence: with a stack the first route found to the goal is not guaranteed to be the shortest.

Marking note: Accept either example. The answer must link FIFO/LIFO to the visiting order, not just name the structures.

C5

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

M4

- (a) BFS counts edges, not costs: S → A → G has 2 edges (the fewest), cost 1 + 8 = 9.
- (b) Settle S (0): A = 1, B = 2. Settle A (1): G = 9. Settle B (2): C = 4. Settle C (4): G = min(9, 6) = 6. Settle G (6). Cheapest route S → B → C → G, cost 6.
- (c) From S the cheapest edge goes to A (1); from A the only new edge goes to G (8): S → A → G, cost 9.
- (d) The walker judges only the next edge. Dijkstra's algorithm always settles the vertex with the smallest total cost from the start; with no negative costs, no later route can make a settled vertex cheaper, so its greedy choices are safe.

★ *Strong answers add that a greedy rule must be checked on small examples, e.g. coins 1, 3, 4 for 6, where greedy fails.*

Marking note: Accept any clear format for the trace (table or list). Full credit needs the correct costs 9 and 6 and a reason in (d) that refers to total cost from the start.

C6 M5

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

- Leaderboard → array: access by index is O(1). Disadvantage: inserting or deleting in the middle requires shifting items, O(n).
- Playlist → linked list (doubly linked is also acceptable): inserting after a known song only changes links, O(1) once the position is known. Disadvantage: reaching position k takes O(n), and each node needs extra memory for its link(s).

★ *Strong answers note that the best structure depends on which operation is most frequent, and may mention that finding the insertion point in a list can still take O(n).*

Marking note: Other justified answers are acceptable (e.g. a dynamic array for a short playlist) if the reasoning uses operation costs correctly.

C7 M7

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

- 6: 6 < 8 go left to 3; 6 > 3 go right; empty, so 6 becomes the right child of 3.
- 9: 9 > 8 go right to 10; 9 < 10 go left; empty, so 9 becomes the left child of 10.
- 14: 14 > 8, 14 > 10, so 14 becomes the right child of 10.
- Inorder traversal (left, node, right): 1, 3, 6, 8, 9, 10, 14, which is sorted order.
- Each comparison sends the search left or right, so it skips a whole subtree. Searching for 9 visits only 8, 10 and 9 (3 nodes) instead of up to 7 values in a list.
- Inserting 1, 3, 6, 8, 9, 10, 14 in increasing order makes every value a right child, so the tree becomes a chain like a linked list; searching for 14 then visits all 7 nodes (O(n)). Self-balancing trees such as AVL trees prevent this.

Marking note: Full credit needs all three insertions placed correctly with reasons, the correct inorder traversal, and the chain problem explained.

C8 M8

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

- (a) $\hat{y} = 0.5 \times 2 + 0 = 1$; $L = (1 - 3)^2 = 4$.
- (b) gradient $= 2 \times (1 - 3) \times 2 = -8$. new $w = 0.5 - 0.1 \times (-8) = 1.3$. The negative gradient means increasing w lowers the loss.
- (c) $\hat{y} = 1.3 \times 2 = 2.6$; $L = (2.6 - 3)^2 = 0.16$, much lower than 4.
- (d) Forward pass: make a prediction. Loss: measure how wrong it is. Backward pass (backpropagation): find how each weight affects the loss. Update: move each weight a small step against its gradient; the learning rate sets the step size.
- One step improves the prediction for one example; repeating over many examples and epochs finds weights that work for the whole dataset. Too large a learning rate can overshoot.

Marking note: Full credit needs the correct numbers (4, −8, 1.3, 2.6, 0.16) and a clear explanation of each step. Accept updating b as well if it is done correctly (gradient −4, b = 0.4).

C9

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

M9

- (a) Each layer computes $Wx + b$; two layers give $W_2(W_1x + b_1) + b_2 = (W_2W_1)x + (W_2b_1 + b_2)$, which is again one linear function. So ten linear layers equal one; they cannot learn XOR or curved boundaries.
- (b) Backpropagation multiplies the gradient by each layer's activation slope. With sigmoid's slope at most 0.25 (and weights near 1), 10 layers shrink it to about $0.25^{10} \approx 10^{-6}$: the vanishing gradient problem.
- (c) Any two of: ReLU in hidden layers; residual connections; good weight initialisation; normalisation.
- (d) House price: linear. Spam: sigmoid. 5 species: softmax. Standard deviation: softplus (must be above 0).

Marking note: Accept a numerical example instead of algebra in (a). Full credit needs both explanations and four correct output activations.

C10

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

M10

- (a) Forecast = week 2: 32, 26, 29, 28, 42. Errors: +1, −1, −2, −1, −3. MAE = 8 / 5 = 1.6. WAPE = 8 / 163 ≈ 4.9%. Bias = −6 / 5 = −1.2 (slightly too low).
- (b) Sales repeat a weekly pattern (Friday highest, Tuesday lowest), which the seasonal naive forecast copies. A new model is only worth using if it beats this simple baseline. The split must follow time (train on the past, test on the future); shuffling leaks future information and makes test scores look too good.
- (c) Shortage cost = 3 − 1 = 2 (lost profit); leftover cost = 1 (waste). Stock to 2 / (2 + 1) = 2 / 3 ≈ 0.67, about the P67. Running out costs twice as much as a leftover, so it pays to stock above the most likely demand.

Marking note: Full credit needs correct numbers in (a) (MAE 1.6, WAPE ≈ 4.9%, bias −1.2), a reasoned baseline and splitting explanation, and the quantile 2/3.

C11

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

M11

- (a) $A \cdot B = 24$, $|A| = |B| = 5$, cosine = 0.96. $A \cdot C = 0$, cosine = 0. B's ratings should count most; C has no overlap with A's taste in these videos.
- (b) Cold start. Use content features (category, length), popular videos, a short questionnaire, or show a mix of videos and learn quickly from clicks.
- (c) Concept drift (sudden). Online learning updates after each new example, so the model adapts quickly; training on a sliding window of recent data gives the new behaviour more influence. Watching the error rate shows when drift happens.
- (d) Users only click what they are shown, so the system shows more of the same: a filter bubble. Mixing in some new items (exploration) helps.

Marking note: Accept other valid cold-start strategies and drift responses (e.g. weighting recent data, retraining when a detector raises an alarm). Full credit needs both cosine values.

C12 M12

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

- (a) Episode 1: Q(2) = 0 + 1 × (1 − 0) = 1; Q(0) and Q(1) stay 0. Episode 2: Q(1) = 0.8 × 1 = 0.8; Q(2) = 1. Episode 3: Q(0) = 0.8 × 0.8 = 0.64; Q(1) = 0.8; Q(2) = 1.
- (b) Each update looks only one step ahead, using the current Q of the next cell. A cell gains value only after the cell after it has some value, so the reward's value moves back one cell per episode, shrinking by γ each step.
- (c) Here the robot was forced to move right. A real agent does not know which action is good, so it must sometimes try actions at random (explore) to discover rewards; ε-greedy acts randomly with probability ε and otherwise takes the best-known action.
- Reward design: agents exploit loopholes, e.g. a boat-racing agent that circles to collect bonus points instead of finishing the race.

Marking note: With α = 1 the new Q equals the target r + γ × Q(next). Full credit needs all nine values in (a) and clear reasoning in (b) and (c).

C13 M12

KEY POINTS · A FULL ANSWER INCLUDES:

‹ BACK

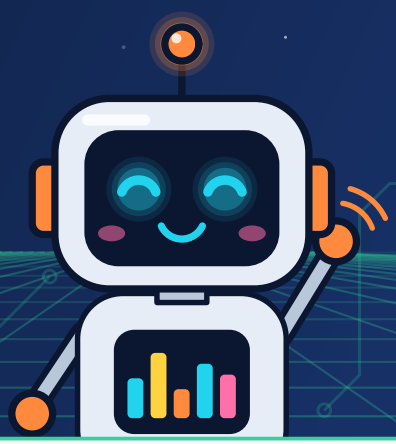
- (1) Overfitting: more data or augmentation, dropout, weight decay or early stopping.
- (2) Underfitting: a bigger model, longer training or better features.
- (3) The learning rate is too high: lower it or clip the gradients.
- (4) A fair guess across 10 classes gives a loss of about ln 10 ≈ 2.30, so 9.7 points to a bug (e.g. labels or the loss set up wrongly). Check the data pipeline first.
- Training set: fits the weights. Validation set: tunes settings and decides when to stop. Test set: one final, honest estimate. Tuning on the test set makes its score too optimistic, because the choices were fitted to it.

Marking note: Accept other reasonable fixes for each case. Full credit needs all four diagnoses and the roles of the three sets.

DEBRIEF IN PROGRESS

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More questions on the next page.



NEXT PAGE >>

REVIEW RECORD

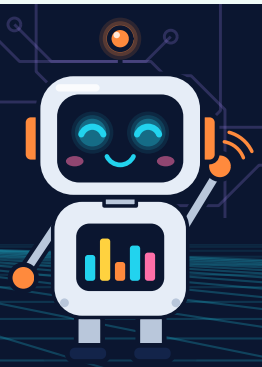
The assignment requires another member to attempt and review every question.

PART	WRITTEN BY	ATTEMPTED & CHECKED BY	DATE	CORRECTIONS MADE
Part A A1–A40	Team	Team		
Part B B1–B42	Team	Team		
Part C C1–C13	Team	Team		

END OF TRANSMISSION

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Review every X, revisit the matching Study Guide mission, then reset and run the test bank again.



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B.O.T.A.K.

Missions in Algorithms, Data Structures and Connectionist AI

TEST BANK V1.0



BUILT BY TEAM B.O.T.A.K.

Nicky Wijaya Suhendra, Devon Reinhart, Darren Tan Siswanto, Neil Edric, Novan Ertanto Vijaya, Glen Randy Erlanda

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